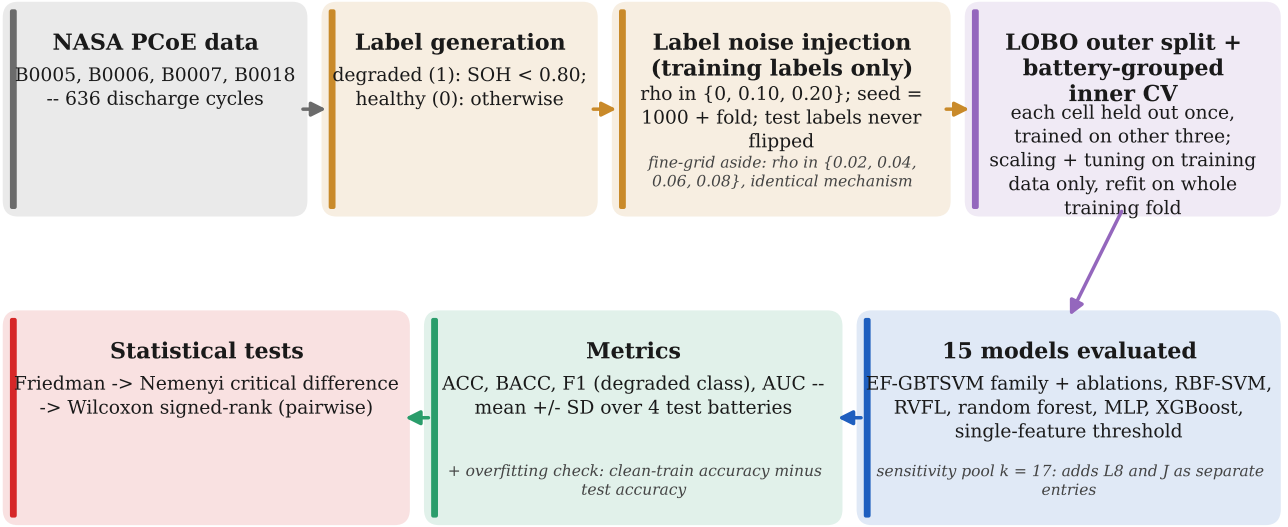
NASA PCoE data
B0005, B0006, B0007, B0018 -- 636 discharge cycles
Label generation
degraded (1): SOH < 0.80; healthy (0): otherwise
Label noise injection (training labels only)
rho in {0, 0.10, 0.20}; seed = 1000 + fold; test labels never flipped
fine-grid aside: rho in {0.02, 0.04, 0.06, 0.08}, identical mechanism
LOBO outer split + battery-grouped inner CV
each cell held out once, trained on other three; scaling + tuning on training data only, refit on whole training fold
15 models evaluated
EF-GBTSVM family + ablations, RBF-SVM, RVFL, random forest, MLP, XGBoost, single-feature threshold
sensitivity pool k = 17: adds L8 and J as separate entries
Metrics
ACC, BACC, F1 (degraded class), AUC -- mean +/- SD over 4 test batteries
+ overfitting check: clean-train accuracy minus test accuracy
Statistical tests
Friedman -> Nemenyi critical difference -> Wilcoxon signed-rank (pairwise)